



Python End-to-End ETL Pipeline Project

This project serves as a comprehensive, hands-on exercise for DataCraft academy students to build an end-to-end Extract, Transform, Load (ETL) pipeline entirely using Python. Students will simulate a real-world data engineering scenario, managing data ingestion from disparate sources, performing rigorous data cleansing and transformation, building a dimensional data model, and generating basic business intelligence (BI) insights.

The project is structured around key milestones, each utilizing Python libraries for data manipulation and analysis.

Project Objectives

1. Master data extraction techniques from different sources (API, Relational Database, Data Lake/Local File System).
2. Implement best practices for data cleaning, standardization, and complex transformations using Python (e.g., Pandas).
3. Design and implement a relational data model (Star Schema or Snowflake) suitable for BI and reporting.
4. Load transformed data into the designed data warehouse model.
5. Derive business insights through data visualization (charting).

Main Milestones

The project will be executed in five distinct phases:

1. Extract Data from Various Sources

This milestone focuses on retrieving raw data from all specified sources and storing it temporarily for processing.

Data Source	Python Library/Technique	Description
API	<code>requests</code>	Connect to an external API to fetch transactional or reference data.
Database Tables	<code>sqlite3</code> or <code>psycopg2/mysql.connector</code>	Connect to and extract data from a relational database (e.g., existing tables).
Data Lake (Local Folders)	<code>os</code> , <code>glob</code> , <code>pandas</code>	Read various file formats (e.g., CSV, JSON, Parquet) from local directory structures mimicking a Data Lake environment.

API Data Source

- <https://openexchangerates.org/>
- Get you app ID

Database Tables

- `order_items`
- `orders`

DataLake

- `brands.csv`
- `categories.csv`
- `customers.csv`
- `products.csv`
- `staffs.csv`
- `stocks.csv`
- `stores.csv`

Data Consolidation: Combine all extracted data into a single CSV file per dataset, enrich them with metadata like extraction timestamp and data source.

2. Data Quality Checks

- **Null Checks:** Identify and handle null values in essential fields.
- **Duplicate Checks:** Detect and remove duplicate rows.
- **Data Validation:** Ensure data types and values are within expected ranges (e.g., price ranges, date limits).
- **Preparation for Staging:** Store cleaned and validated data in 'staging_1' folder for transformation.

3. Transformation

This phase involves cleaning, standardizing, and applying necessary business logic to the extracted raw data.

Key Transformation Tasks:

- **Currency Conversion:** Merge latest currency exchange rates to calculate local prices.
- **Delivery Metrics:** Add columns to track delivery performance, such as late deliveries and latency days.
- **Locality Flag:** Determine if customers are local based on proximity to stores.
- **Lookup Tables:** Resolve ambiguous columns by creating and utilizing lookup tables for order statuses.
- **Transformed Data Staging:** Output transformed data to 'staging_2' for further processing.

4. Build a Data Model

Students will design and implement a dimensional model that supports efficient querying for reporting.

Component	Description
Fact Tables	Contains quantitative measures and foreign keys to dimension tables.
Dimension Tables	Contains descriptive attributes for fact data (e.g., Date, Customer, Product).
Schema Design	Defining the relationships and structure (e.g., Star Schema).

- **Data Merging:** Integrate orders, items, and product details into a unified dataset for deeper analysis.
- **Visualization Creation:** Develop at least three types of visualizations to illustrate key metrics and trends.
- **Documentation:** Document the modeling techniques and visualization choices.
- **Result Compilation:** Organize final datasets and visualizations in 'Information Mart' and 'Visualization' folders respectively.

5. Reporting using Charts

The final step is to leverage the structured data model to generate meaningful business insights using data visualization.

- **Library:** `matplotlib` or `seaborn` (and optionally `pandas.plotting`).
 - **Required Insights (Examples):**
 - Time-series analysis (e.g., Sales Over Time).
 - Top N performance (e.g., Best Selling Products).
 - Distribution analysis (e.g., Customer Segmentation).
 - Geographical visualization (if location data is available).
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Project Deliverables

Each student must provide the following:

1. **Source Code:** A well-documented Python project containing all scripts for the ETL pipeline.
2. **Schema Diagram:** A visual representation of the implemented data model (Fact and Dimension tables).
3. **Final Report:** A document detailing the steps taken, challenges encountered, and a summary of the data insights found. This report should include the generated charts.

Recommended Python Libraries

Students are encouraged to use the following libraries for their respective tasks:

- **Data Extraction & Web Interaction:** `requests`
- **Database Interaction:** `sqlite3`, `SQLAlchemy` (optional, for advanced students)
- **Data Manipulation & Transformation:** `pandas`, `numpy`
- **Data Visualization:** `matplotlib`, `seaborn`